

KYLE SANTOS

Embedded Systems and AI Computer Vision Engineer

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PROFESSIONAL SUMMARY

Electronics Engineering graduate specializing in embedded systems and AI-based computer vision. Experienced in designing and integrating hardware-software systems using ESP32, Arduino, and Raspberry Pi, with applied knowledge in YOLO-based object detection. Developed real-world engineering solutions focused on automation and environmental monitoring, including an AI-powered microplastic detection system for retail and domestic rock salt screening. Comfortable working across circuit design, sensors, camera-based inspection, prototype testing, user interfaces, and technical documentation.

EDUCATION

Bachelor of Science in Electronics Engineering Graduated July 2026
Marikina Polytechnic College
Capstone: Development of an Automated Optical Detector for Microplastics in Rock Salt for Retail and Domestic Use
Relevant Coursework: Embedded Systems, Digital Electronics, Computer Networks, Signal Processing, Electronics Design

TECHNICAL SKILLS

- Programming:** Python, OpenCV, YOLOv11, C/C++, Arduino/ESP32 firmware logic
- Embedded Systems:** ESP32, Arduino UNO, Raspberry Pi 5, GPIO, sensor integration, motor control
- Computer Vision and AI:** object detection, dataset annotation, model training, validation metrics, detection UI
- Coding Platforms and AI Workflow:** VS Code, Antigravity coding platform, ChatGPT, Claude, prompt-guided debugging, documentation, code review, workflow planning
- API and Data Handling:** API concepts, JSON, HTTP requests, reading documentation, integrating AI/API-backed features
- Electronics and Hardware:** circuit design, MOSFET switching, TP4056 battery systems, DC-DC converters, LED control
- Interfaces and Output:** LCD/OLED displays, touchscreen workflows, report generation, graph and analysis views
- Tools and Simulation:** VS Code, Arduino IDE, Wokwi, Tinkercad, Cisco Packet Tracer, EasyEDA, Canva

ENGINEERING PROJECT EXPERIENCE

Capstone Project Lead and Full-System Developer | MICROTTECT Automated Optical Detector Project Lead | April 2026
Developed an AI-integrated optical detection system for identifying, classifying, and quantifying microplastic particles in rock salt for retail and domestic use. The project addressed the time, cost, and repeatability limitations of manual microscopy by combining optical imaging, embedded processing, AI-assisted analysis, frontend result UI, backend processing logic, UI/UX flow, and report generation.

- Integrated Raspberry Pi 5, HQ camera module, illumination system, display interface, and prototype enclosure.
- Implemented a YOLOv11 object detection workflow with OpenCV image handling and result visualization.
- Developed the operator-facing interface and workflow from sample capture to result review, graph views, and report output.
- Generated analytical outputs including particle count, estimated particle size, MP/g, confidence, and contamination level.
- Used ADDIE methodology and ISO/IEC 25010 evaluation criteria for development and assessment.
- Reported 69.71% mean detection consistency, with precision 0.826, recall 0.733, and mAP@50 of 0.814.

MICROTTECT HARDWARE SCOPE

- Raspberry Pi 5 processing unit with 64GB SD storage and Pi power supply.
- HQ camera module, camera stand, LED lamp/ring light, and sample viewing area.
- 7-inch LCD monitor, sampling containers, Petri dishes, labels, and enclosed prototype body.

MICROTTECT SOFTWARE SCOPE

- Image capture and preprocessing for sample inspection.
- Bounding-box detection with pixel-to-micrometer size estimation.
- Detection dashboard with histogram/line analysis and PDF-style reporting flow.

ADDITIONAL ENGINEERING PROJECTS

ESP32-Based Remote Monitoring System

Embedded Systems

- Built a real-time monitoring system integrating motion, temperature, and smoke sensors.
- Displayed environmental readings and system status using an LCD interface.
- Developed modular testing and validation for each sensor component before integration.
- Practiced threshold-based decision logic for environment status monitoring and debugging.

Smart Pet Feeder System

Automation Prototype

- Designed an automated feeding mechanism using ultrasonic and LDR sensors.
- Controlled servo motors for dispensing and integrated buzzer alerts for notifications.
- Implemented OLED display feedback for system status and operation visibility.
- Coordinated sensing, actuation, timing, and user feedback in one physical automation workflow.

Autonomous Line-Following and Maze-Solving Robot

Robotics

- Developed autonomous navigation using IR sensors, ultrasonic sensing, and motor control.
- Implemented left-hand and right-hand rule algorithms for maze solving.
- Integrated Bluetooth control through MIT App Inventor for manual override functionality.
- Tuned sensor thresholds and navigation behavior to improve path following and obstacle response.

App Projects | HerCare and Tamar

Product and Frontend Development

- Built app experiences that complement my embedded systems work with user-facing software and product thinking.
- Used calm interface structure, practical workflows, and reusable screens to make the apps easier to use.
- Expanded the portfolio beyond hardware by showing both technical implementation and app-focused UX decisions.

RESEARCH, TESTING, AND TECHNICAL DOCUMENTATION

- Prepared capstone documentation covering problem background, objectives, scope, related literature, methodology, results, conclusions, and recommendations.
- Conducted controlled testing of rock salt samples across multiple sample sources and trials.
- Used percentage, mean, and weighted average treatments to summarize system performance and user evaluation data.
- Documented prototype visuals, circuit diagrams, 3D design, hardware development, object detection training, and UI outputs.
- Used AI tools to support research organization, technical writing review, debugging direction, and clearer project documentation.
- Identified future improvements such as expanded datasets, stronger laboratory validation, and improved imaging/illumination components.

AFFILIATIONS

- Institute of Electronics Engineers of the Philippines, Student Member
- Electronics Engineering Student Society, Member

LANGUAGES AND INTERESTS

- **Languages:** Filipino and English.
- **Technical Interests:** AI usage, API integrations, embedded systems, electronics prototyping, automation, troubleshooting.
- **Working Style:** hands-on prototyping, careful documentation, iterative testing, and learning new engineering tools.

ADDITIONAL INFORMATION

- Experience in system prototyping, testing, debugging, sensor integration, and technical documentation.
- Strong focus on integrating AI with physical hardware systems for practical, real-world applications.
- Comfortable using AI assistants responsibly as learning, planning, debugging, and documentation tools while validating final technical decisions.
- Portfolio includes project case studies, final paper, 3D design renders, and detection UI screenshots.